Technical Specifications

Confocal Microscope – Zeiss LSM 510 *DUO Vario*

This versatile laser scanning confocal microscope has 3 confocal platforms and 9 laser lines for excitation from near UV to far red (see Table).

The *LSM 5* point-scanning system delivers true 5D experimental capability (X,Y,Z,Time and λ) creating detailed Z-stack images for precise 3D rendering and time series acquisition.

The *LSM 5 Live* adds a beam scanning confocal head that allows for ultra 'high-speed' capture of live cell processes, tracking of biological molecules and physiological processes through photobleaching, photoactivation, photoconversion, and uncaging, employing confocal quality imaging at speeds up to 120 fps (512x512 resolution).

The *LSM 5 Meta* can separate and acquire 32 spectral channels which allows for the precise separation of fluorochromes. It can completely eliminate crosstalk and auto fluorescence in difficult samples; perform emission fingerprinting; or characterize unknown fluorescence signals through spectral analysis

Objectives **Note some lenses used on AxioImager Z1 can be used on this scope as well**	5x/ 0.16 PH1 EC Plan Neofluar 10x/0.3 Plan Neofluar 20x/0.8 Plan Apochromat 40x/1.3 Oil Plan Apochromat 50x /0.55 DIC EC Epiplan-Neofluar 63x/1.3 ImmKor DIC LCI Plan Neofluar 63x/1.4 Oil DIC Plan Apochromat	
LSM Laser Lines for Excitation	100mW 488 – 100 Diode (for live Cell Imaging) 30mW 458, 477, 488, 514 Argon/2 Ion Laser (for standard LSM) 1mW 543nm HeNe (for standard LSM) 5mW 633nm HeNe (for standard LSM) 50mW 405 – 50 Diode (for live cell and standard LSM) 75mW 532 – 75 DPSS (for live cell imaging)	
LSM Emission Filters	**Channel 2** LP420 LP475 LP505 BP 575-630 IR BP 420-480 BP 465-510 IR BP 505-530 BP 505-575 IR	**Channel 3** LP 505 LP 560 LP 615 LP 650 BP 505-530 BP 505-550 BP 520-555 IR BP 560-615 IR
LIVE Emission Filters	**Channel 1** LP 420 LP 505 BP 495-520+BP 550-615 IR LP 550 BP 560-675 LP 650 BP 665-750 IR 0 BP 665-750 IR 90	**Channel 2** BP 415-480 BP 445-525 BP 500-525 BP 550-615 BP 505-610 IR